

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

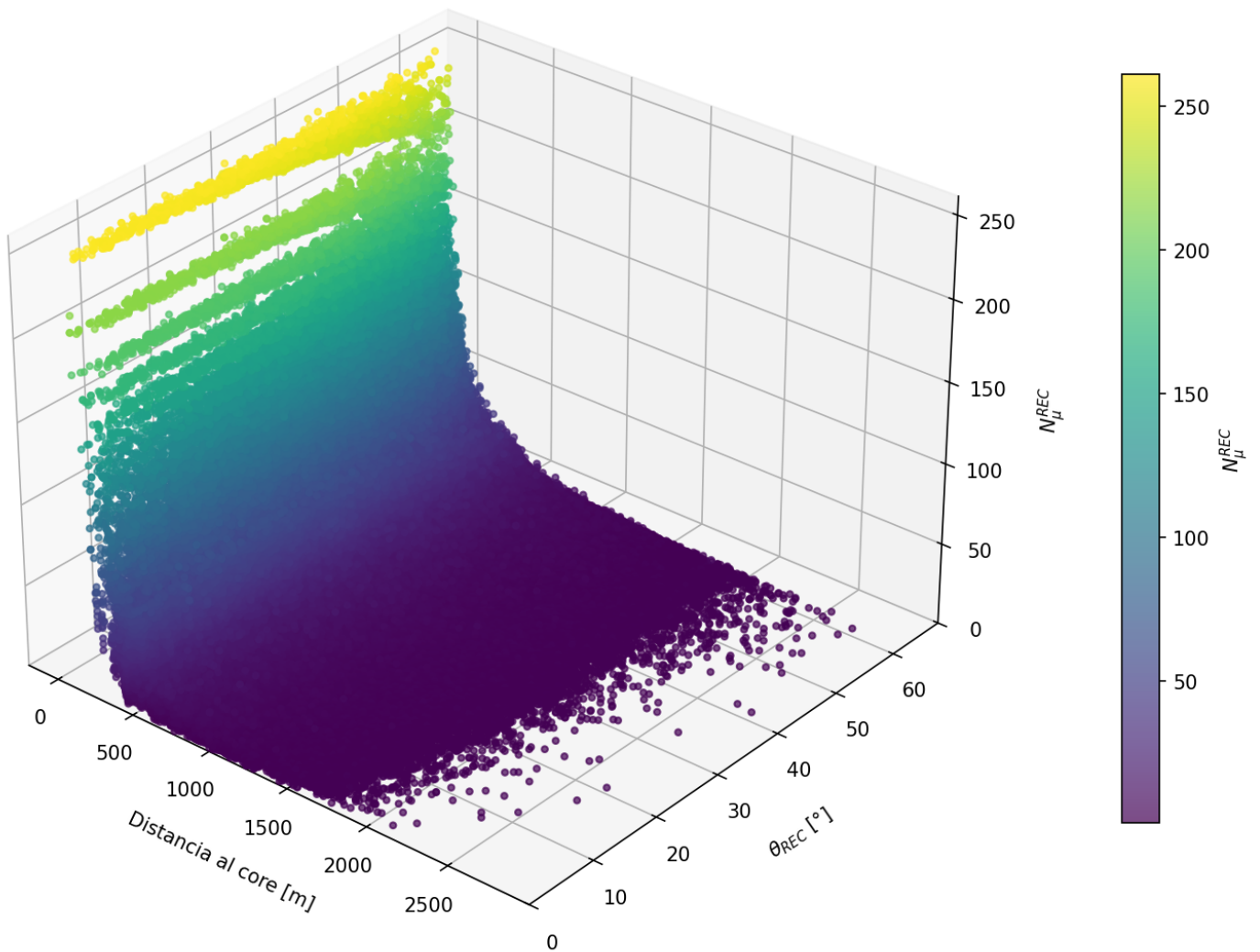
Parámetro	Valor
Modelo Hadrónico	EPOS-LHC
Primario	Helio
Rango Energía	18.0 - 18.5 log(eV)
Carpeta	
Eventos Totales	5,000
Módulos Totales	2,227,440

1. Limpieza de Datos

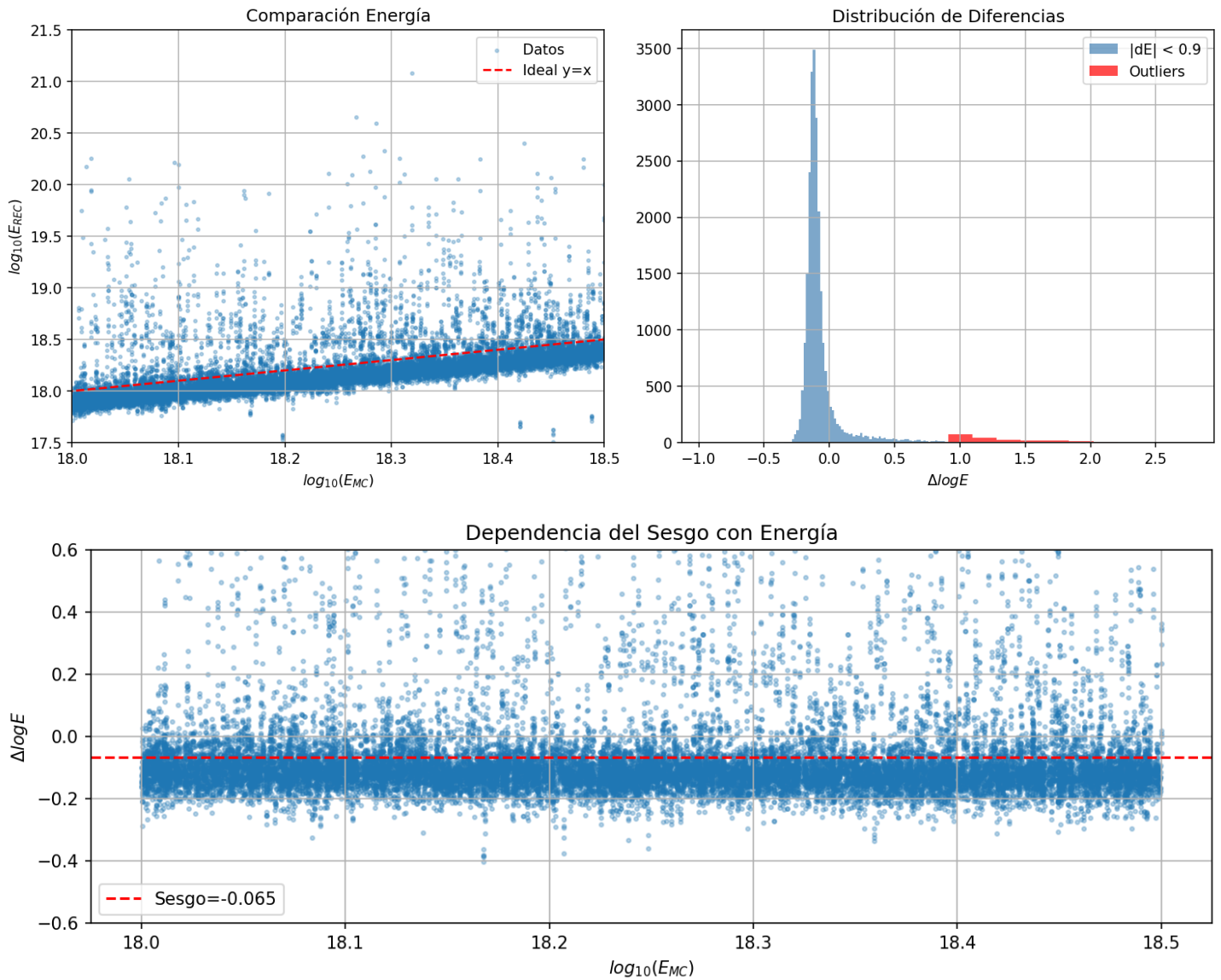
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	2227440
Filas Eliminadas	109164 (4.90%)
Filas Finales	2118276
Causa Principal	NaN en logE_REC (109164)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

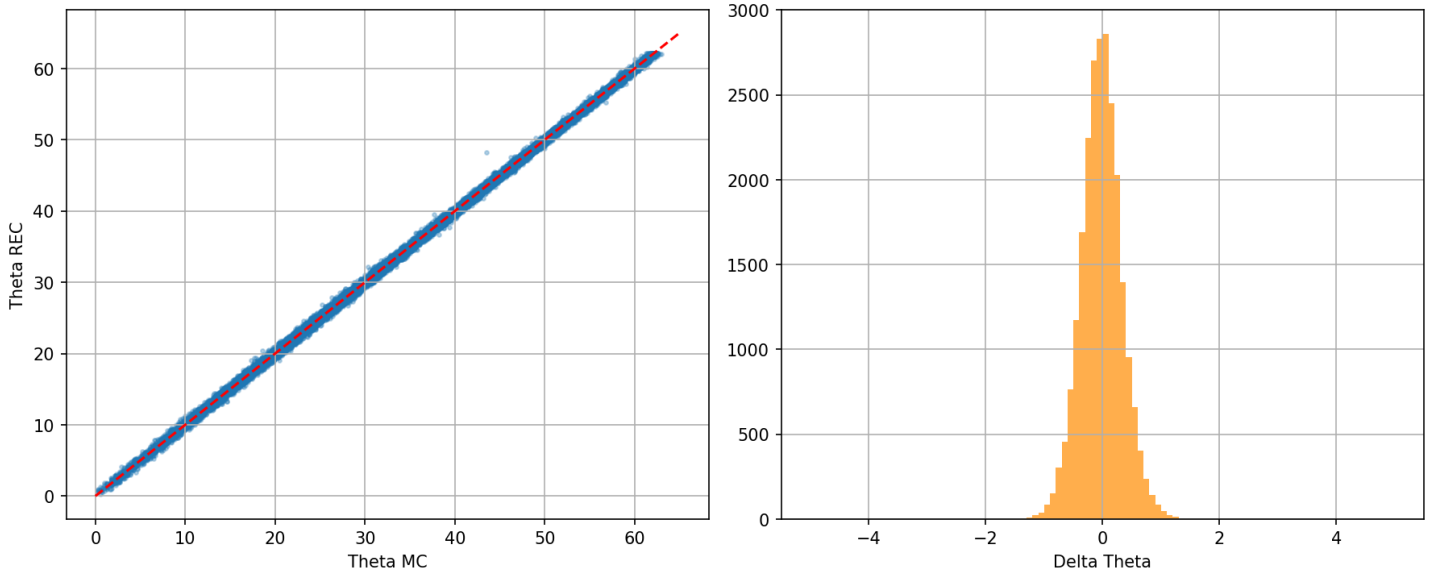
3. Resolución de Energía



Sesgo (Mean): -0.0654
 Resolución (Std): 0.1941
 Outliers (> 0.9): 200 (0.84%)

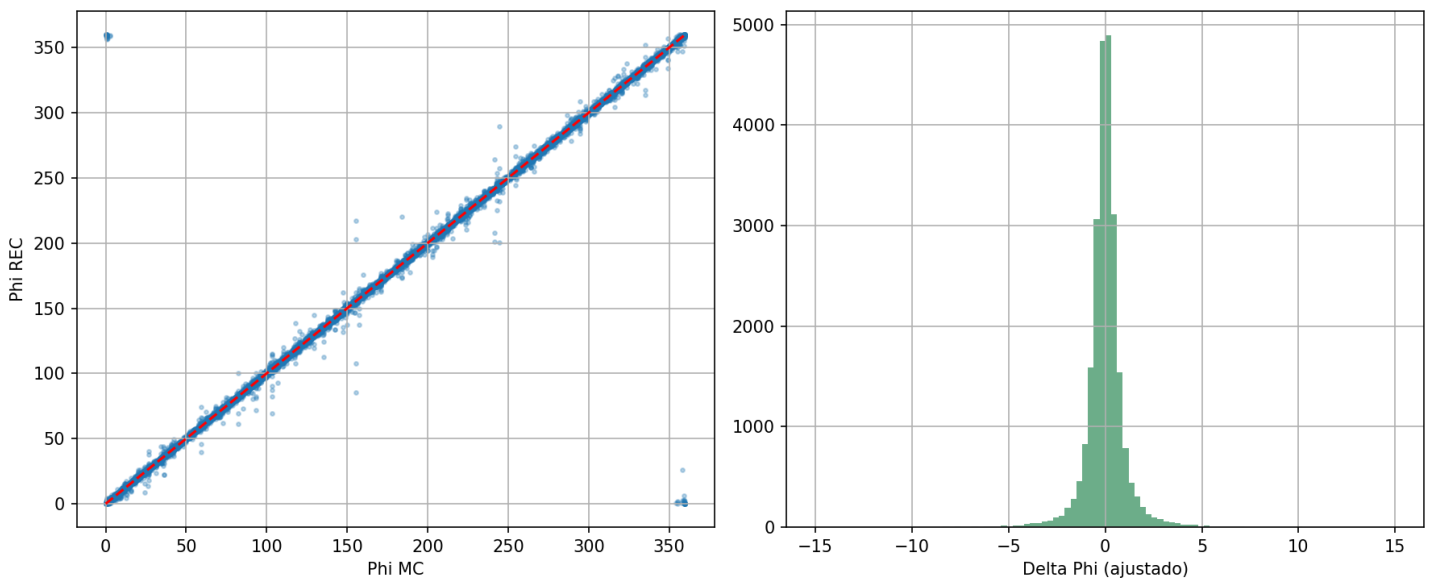
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0168 deg
Resolución: 0.3504 deg
Outliers ($> 10.0^\circ$): 0 (0.00%)

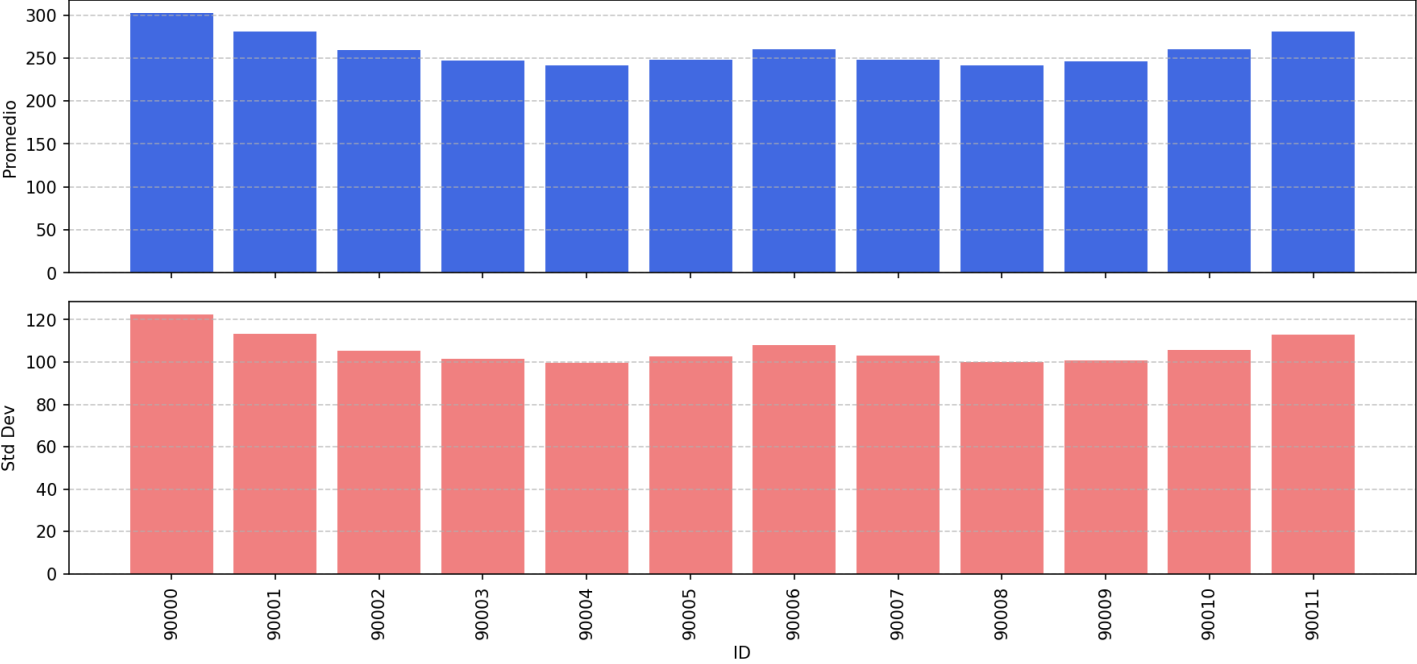
Phi (Azimutal)



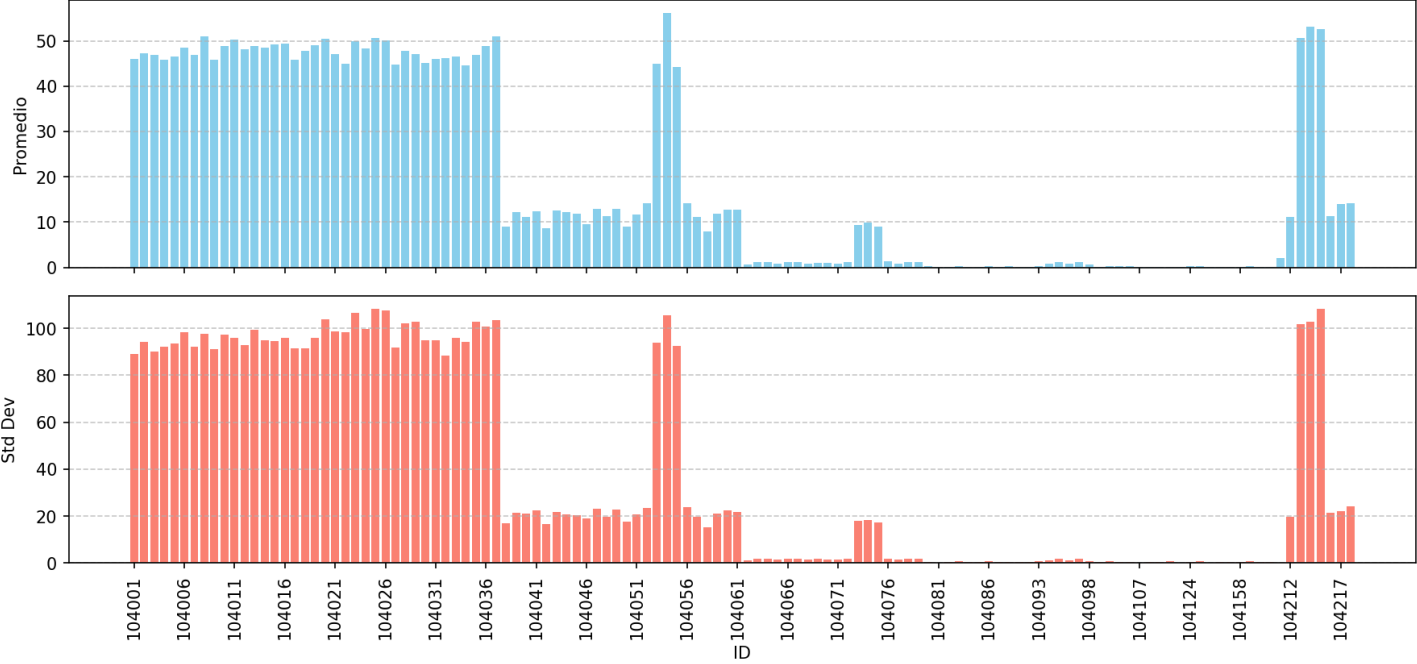
Sesgo (Median): 0.0011 deg
Resolución (MAD): 0.3865 deg
Outliers ($> 15.0^\circ$): 32 (0.13%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

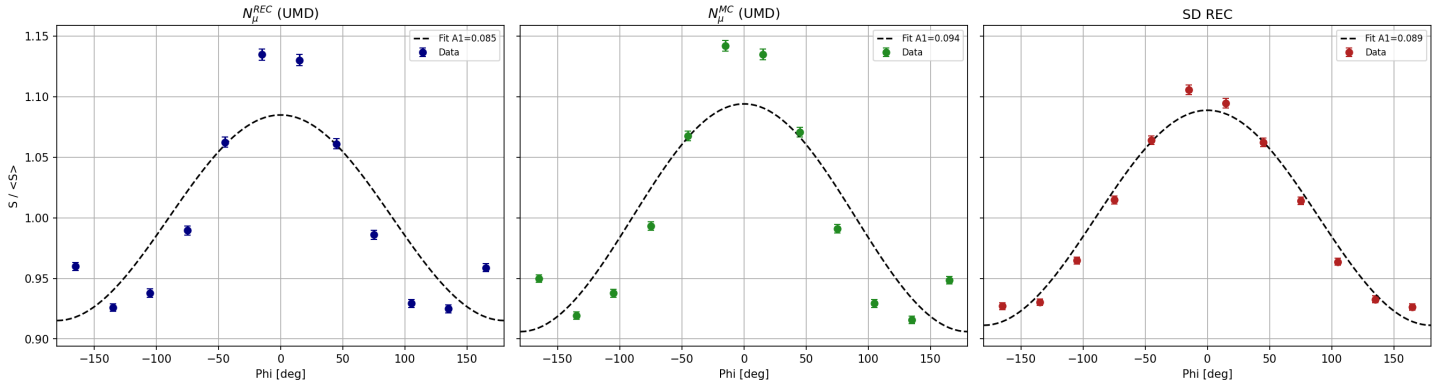
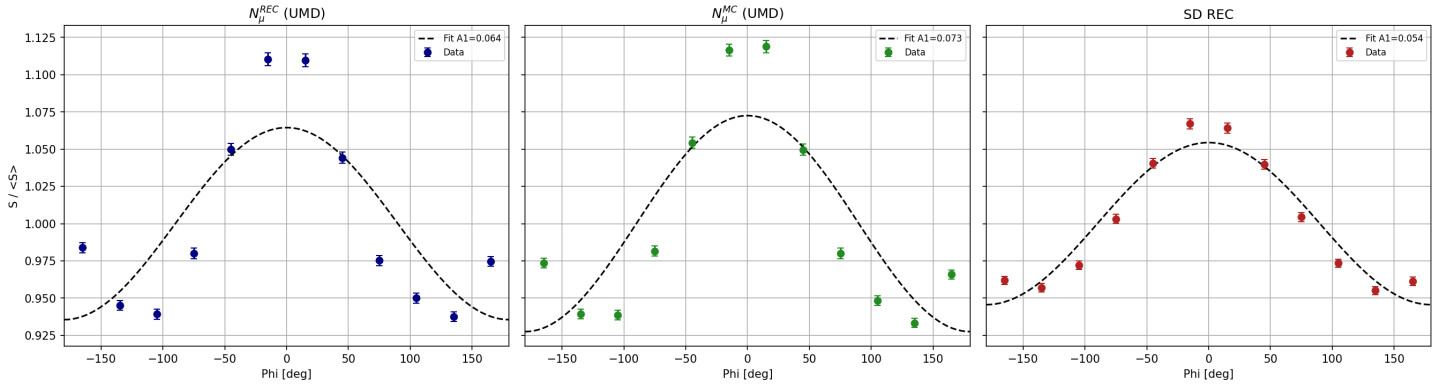
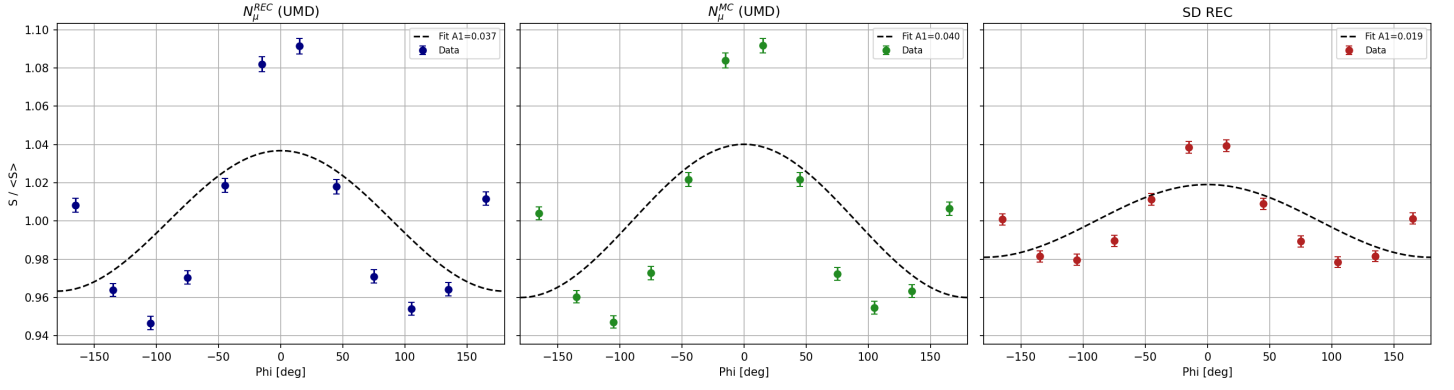


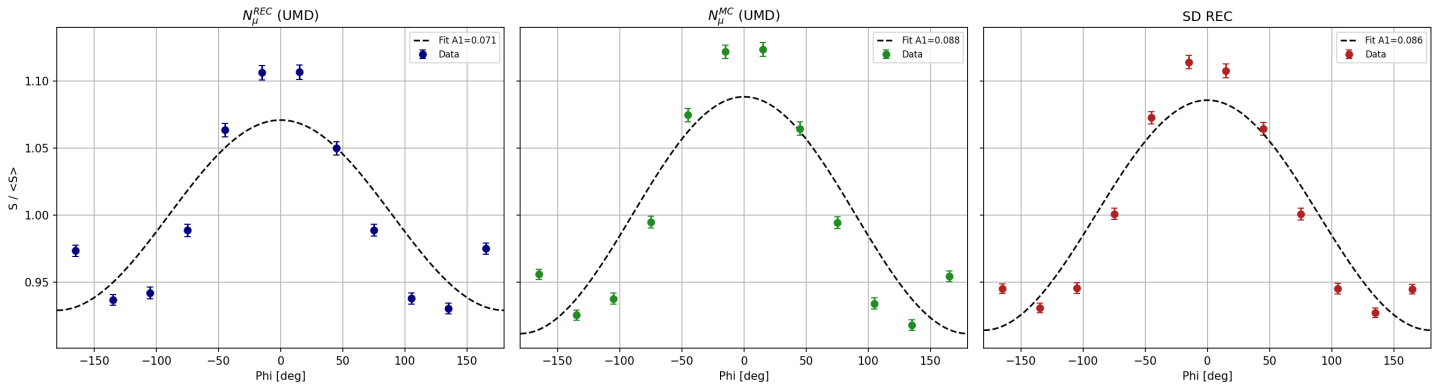
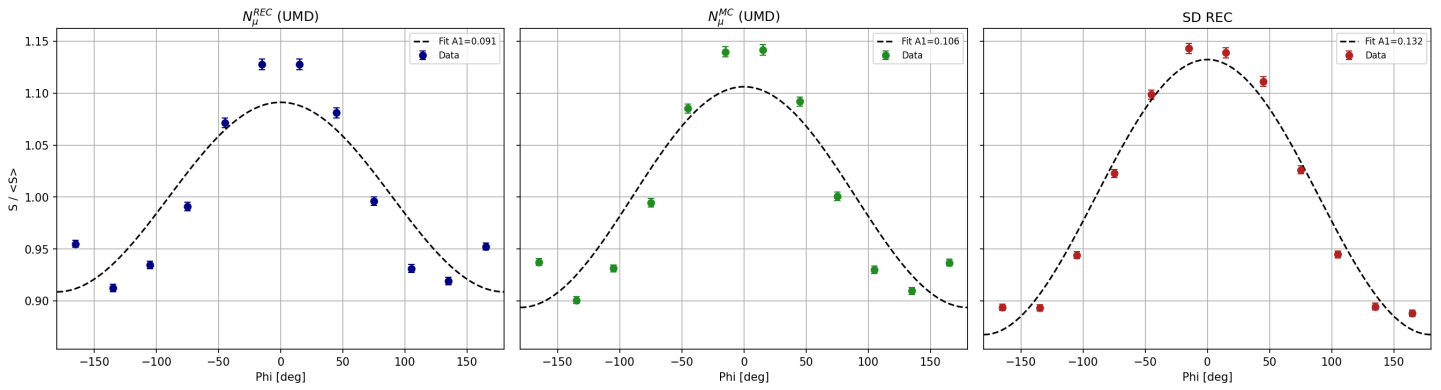
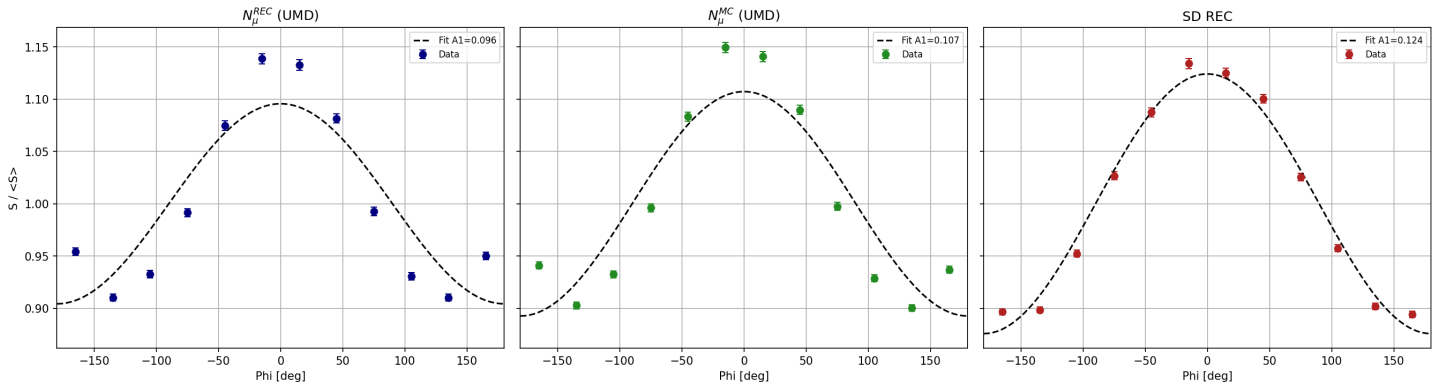
Uniformidad: Infill (104k)



6. Asimetría Azimutal

Bin 0: Theta 0.3 - 21.2 deg





7. Resumen de Ajustes (Comparación)

Resultados Detallados: Amplitud A1 y Error

theta_range	A1_UMD_REC	Err_UMD_REC	A1_UMD_MC	Err_UMD_MC	A1_SD_REC	Err_SD_REC
0.3-21.2	0.0367	0.0157	0.0400	0.0152	0.0190	0.0064
21.2-30.7	0.0645	0.0150	0.0725	0.0145	0.0544	0.0046
30.7-38.7	0.0849	0.0152	0.0940	0.0145	0.0888	0.0044
38.7-46.2	0.0957	0.0154	0.1073	0.0148	0.1242	0.0051
46.2-53.8	0.0913	0.0146	0.1063	0.0141	0.1324	0.0068
53.8-62.2	0.0709	0.0141	0.0883	0.0137	0.0858	0.0101

